

A Comparative Analysis of K-Means and GMM Clustering Post PCA, ICA, RP, and SVD Feature Transformation

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I. EDA ON DATASETS

Exploratory data analysis (EDA) of the Adult income dataset shows 45,222 data points with 15 features. These are a mix of categorical variables (workclass, occupation, marital-status etc.) and continuous variables (age, capital-gain/capital-loss, hours-per-week, etc.). Some strong financial indicators like capital-gain and capital-loss, show an extreme right skew and sparsity with around 75% of the reported data being 0. Due to this sparsity, the model must often rely on other less reliable features such as age and education to predict income class. Many categorical variables have imbalanced frequencies, examples being white in race (86%) and United States in native-country. Additionally, several categorical features contain many low-frequency categories, resulting in sparse high-dimensional representations after encoding. The target variable (income) also has a strong class imbalance with approximately a 3:1 ratio between $\leq 50K$ vs $> 50K$ income. Income patterns appear to arise from conditional interactions between features such as education, occupation, and age rather than linear relationships.

EDA on the Wine Quality dataset shows us that the dataset has 6497 wines with 11 features. The class and quality features are identical. There are no missing values. Aside from type, all of the features are continuous variables. Several of the features are right-skewed. Some examples shown through diagrams are residual_sugar, free_sulfur_dioxide, and total_sulfur_dioxide. The features free_sulfur_dioxide and total_sulfur_dioxide are strongly correlated. Alcohol is strongly negatively correlated with density, and there are moderate associations with quality as higher alcohol and sulphates tend to align with better quality. Higher volatile_acidity and type (1 - white wine, 0 - red wine) tend to align with lower quality. Multicollinearity between other features in the dataset is present but not extreme beyond these. Most feature-quality correlations are moderate between 0.3-0.4 and highly collinear groups are only weakly correlated with quality. There seems to be a large variance due to human perception, noise and unobserved variables. The class balance of the target variable (wine) is skewed towards mid-scored (3-5). Due to this, accuracy is less informative and F1 score will be more useful in assessing performance of the models.

II. HYPOTHESES

On the Adult dataset, clustering is expected to produce low silhouette scores for both K-Means and GMM due to the high dimensionality, sparsity from one-hot encoding, and class imbalance that causes natural clusters to be dominated by

demographic subgroups rather than income classes. K-Means is expected to struggle more than GMM given its reliance on Euclidean distance and spherical cluster assumptions, which break down in sparse high-dimensional spaces. For dimensionality reduction, PCA and RP are expected to be the most beneficial. PCA eliminates the collinear redundancy caused by encoding, and RP preserves pairwise distances in the compressed space. ICA is expected to be less stable due to the sparse binary nature of the features. Dimensionality reduction is expected to improve cluster structure but not produce clean income-class separation.

On the Wine dataset, clustering is expected to identify moderate geometric structure which corresponds to chemical composition rather than quality labels. Silhouette scores are expected to be moderate on K-Means. GMM may perform better (higher macro-F1 on NN) but have lower silhouette scores since it models clusters probabilistically and can capture overlapping distributions that arise from conditional interactions. For dimensionality reduction, PCA is expected to be the most effective dimensionality reduction method due to the dataset's correlated continuous features. ICA may identify hidden chemical factors but this comes at the cost of stability as feature correlations violate independence assumptions. RP is expected to offer minimal benefit given the already low dimensionality of the dataset. Neural network performance (F1-Macro) on reduced dimension datasets is not expected to surpass the SL report baseline as Wine quality predictions are limited by label subjectivity and moderate feature-quality correlations that cannot be resolved by reduced dimensions. Any improvement in F1-Macro from cluster derived features on the neural network is expected to be marginal, as the cluster structure is expected to reveal Wine type rather than Wine quality due to the limited correlation between features and quality seen in previous reports.

III. METHODS

In step 1, the structure of the Wine and Adult datasets is changed by applying two unsupervised learning methods, K-Means and Expectation-Maximization via Gaussian Mixture Models (GMMs), directly to the raw feature space. For each dataset, a consistent preprocessing pipeline was applied. Categorical features were transformed using one-hot encoding and numerical features were standardized using StandardScaler to place all variables on a comparable scale prior to distance-based clustering. After preprocessing, two clustering analysis were conducted for each Adult dataset and Wine dataset. For K-means, a sweep over cluster counts

$k = \{2, 3, \dots, 10\}$, using $n_{\text{init}}=10$ independent random initializations and a maximum of 300 iterations per run to avoid convergence to poor local minima. For each value of k , the inertia (sum of squared distances to nearest centroid) and the silhouette coefficient were computed on up to 3000 samples to ensure scalability. The optimal number of clusters was selected as the k that yielded the highest silhouette score, and a final K-Means model was refitted using this choice. For the GMM/EM analysis, we similarly evaluated models with $k = \{2, 3, \dots, 10\}$ mixture components, using full-covariance Gaussians, $n_{\text{init}}=3$ initializations, and a maximum of 200 EM iterations. For each model the BIC, AIC, and silhouette coefficient were computed. The optimal value of k was chosen as the one minimizing the BIC score, as this reflects the best trade-off between model-fit and complexity. For both clustering algorithms, the chosen k was compared to the true number of classes (provided for evaluation not model fitting) and cluster quality was assessed through silhouette score and comparison against true labels.

In step 2, the methodology for dimensionality reduction involved applying Principal Component Analysis (PCA), Independent Component Analysis (ICA) and Sparse Random Projection (RP) to each preprocessed dataset to create and compare lower-dimensional representations. For PCA, the number of components was required to capture 95% of the original data's variance, establishing a target dimensionality that was then used for both ICA and RP to ensure a fair comparison. An Eigenvalue Distribution Analysis was also performed to determine the optimal number of components and assess the variance structure of each dataset. The ICA transformation was implemented using FastICA algorithm, which includes an internal whitening step, and the statistical independence of its resulting components was evaluated by measuring their kurtosis. For RP, a sparse random matrix was used to project the data, as this is efficient and aims to preserve the geometric structure of the data. RP was additionally run 10 times with different seeds with K-Means clustering applied to check for instability. Finally, the transformed spaces were visualized by creating 2D scatter plots of the first two dimensions of each method. A controlled noise sensitivity experiment that injects varying levels of Gaussian noise (σ) was performed to measure how effectively each DR method preserves cluster separation under degradation.

In step 3, dimensionality reduction methods were each tested on both clustering methods (K-Means and GMM) on both datasets. First, each dataset had baseline clustering in its raw feature space where the clusters (k) were identified separately for K-Means and GMM. Their corresponding silhouette scores were recorded. For GMM, BIC and Log-likelihood per Sample (LLPS) values were recorded. Next, PCA, ICA, and RP were applied to each dataset to obtain lower-dimensional space. For each method, the transformed data and resulting number of retained components were stored. Finally, the clustering algorithms were applied to each reduced dataset using the same optimal k determined in the baseline step. For both K-Means and GMM, the silhouette scores were computed, as was BIC for GMM.

The change in scores relative to the baseline were computed. The GMM results on both datasets were analyzed using LLPS as well. This is because BIC's absolute value changes drastically depending on the dimensionality and scaling of the reduced space. Additionally, Silhouette scores measure spatial separation and "roundness," which favors K-Means. For these reasons, instead of just using silhouette score, LLPS is also used to compare the GMM models against the baseline. The hardware class used for all experiments was CPU: Intel(R) Xeon(R) CPU @ 2.20GHz, OS Class: Ubuntu 22.04.5 LTS.

IV. CLUSTERING RESULTS

The clustering results (Figure 1) largely align with the hypothesis on Adult established after EDA. There was an expected lack of clear separability. As predicted, both K-Means and GMM performed poorly on Adult, with silhouette scores of only 0.1241 (K-Means) and 0.0281 (GMM), confirming that the dataset does not exhibit well-defined clusters under Gaussian or Euclidean assumptions. The optimal $k=4$ for KMeans and $k=7$ for GMM do not align with the true two-class income labels. When GMM is evaluated at the same $k=4$ as K-Means, it has a score of 0.0384, much lower than K-Means. This contradicts the hypothesis that GMM would perform better. The high dimensionality of the dataset probably degrades the GMM performance more than expected. The PCA projection graph for K-Means shows us that three of the four clusters form the large mass that dominates the plot while one cluster (red) sits in a separate region. The true label plot shows us that dominant color throughout is blue ($<50k$) with only a modest sprinkling of cyan ($\geq 50k$) that does not cluster into any distinct region. K-Means seems to have found subgroups with particular demographic features rather than income classes. This is consistent with the hypothesis that income arises from conditional interactions between features that cannot be simply recovered geometrically. GMM plot shows us that the seven components blend across the entire cloud with no boundaries. GMM's components appear to be local density variations rather than meaningful demographic subgroups. The silhouette curve scores fluctuate from 0.055 at $k=2$ down to 0.039 at $k=4$, then spike irregularly. This pattern reflects unstable EM convergence and overlapping mixture components. Silhouette values never exceed 0.09, showing extremely poor separation. The instability directly results from the 104-dimensional sparse space, where Gaussian covariance estimates become noisy and components overlap heavily. To improve clustering on Adult, dimensionality reduction before clustering would be helpful.

The Wine dataset shows some clusterability, in line with the hypothesis. K-means achieved a silhouette score of 0.3366 at $k=2$, this is higher than any configuration on the Adult dataset. The optimal k is much smaller than the 8 classes, this tells us that as predicted, the wine quality is only loosely connected to chemical composition. This is confirmed in the PCA projection, two clusters correspond to the two distinct lobes in the scatter plot. The true label plot

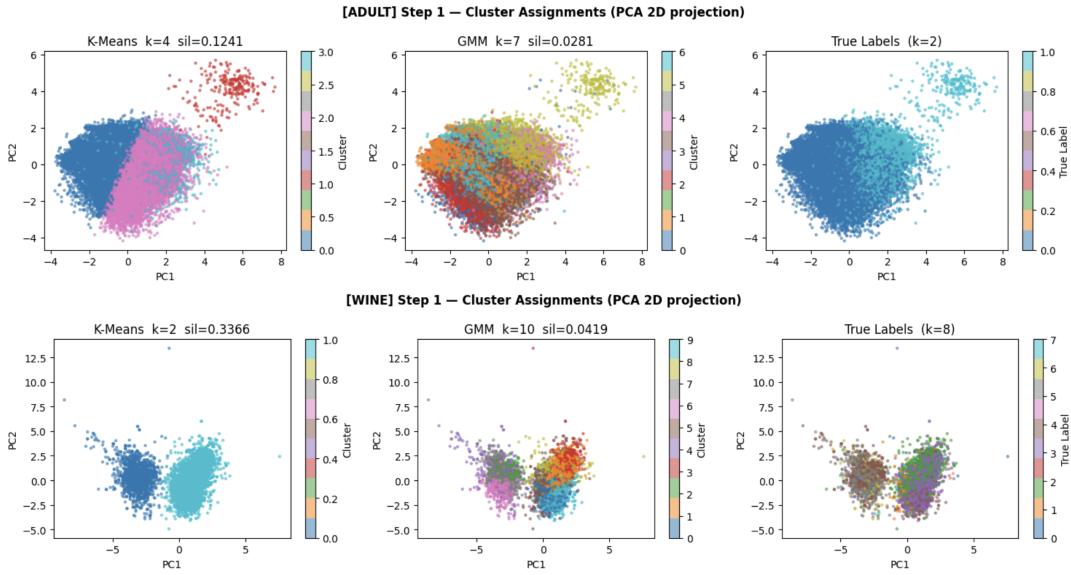


Fig. 1: Cluster Assignments on Adult and Wine (Raw Data)

shows us that the labels are distributed completely randomly across both chemical lobes with no spatial tendency to separate by quality. The GMM results have $k = 10$ at minimum BIC, the silhouette score dropped to 0.0419 here. This indicates additional gaussian components probably fit local density rather than meaningful clusters. This is seen in the scatter plot where ten colors are randomly distributed across the plot with slightly overlapping boundaries. At the shared $k=2$, GMM achieves a silhouette of 0.3346 versus K-Means' 0.3366 (a difference of just 0.002) confirming that both algorithms find essentially the same binary chemical structure when constrained to the same k , and that the apparent performance gap between them in the primary results is entirely due to GMM selecting $k=10$ via BIC.

V. DIMENSIONALITY REDUCTION

The hypothesis correctly predicted that PCA and Random Projection would be the most beneficial for Adult as seen in Figure 2. PCA reduces the dimensions from 104 to 31 while retaining 95.2% of variance, this is the most impactful transformation across both datasets. Adult eigenvalue distribution shows a sharp drop in the first 7 components followed by a long flat tail of 24 near-zero components. This confirms the hypothesis that one-hot encoding produces a bloated, redundant feature space as most of the space was noise. The compression restores the geometric meaningfulness to Euclidean distance. The PCA plot shows effective separation among the three methods. A large, dense cluster is visibly distinguished from a smaller, more diffuse group, which likely corresponds to the imbalanced $>50K$ and $\leq 50K$ income classes. Clustering is expected to perform well on this space as the variance is dense and there is less noise. This will probably result in better classification.

ICA has a mean kurtosis of 10.41 which is well above Gaussian. The heavy tails arise from the sparse nature of the underlying features rather than meaningful structure. The

hypothesis that ICA would be unstable on this dataset was confirmed by the scatter plot. The plot is dominated by distinct horizontal bands. This occurs because ICA isolates the individual sparse vectors from the one-hot encoded categorical features (e.g., a vector representing a single occupation). The plot suggests that ICA is not suitable for this type of data, as it produces a mathematically "independent" but useless and artificial structure. Clustering might be difficult on this space as K-means assumes spherical clusters, although, GMM might perform better but might just end up clustering the outliers.

RP's scatter plot shows a grid like structure. This is the projection of the original 104-dimension space. Points from both classes are intermingled across the lines. The algorithm captures the structure of projection of the encoded features. RP has good reconstruction quality. The Johnson-Lindenstrauss lemma [1] guarantees approximate preservation of pairwise Euclidean distances under random projections, thereby maintaining the geometric structure required for distance-based clustering. Adult is stable on RP when run multiple times (10 seeds) with a standard deviation of 0.0194 (silhouette score on K-Means) and range of 0.0735, three times tighter than wine. This is because with 104 dimensions compressed to 31 and 45,000 data points, RP averages over many feature activations per data point.

The rank of the data is 96 out of 104 with a ratio of 0.923. The redundancy of 8 dimensions is due to one-hot encoding. The condition number of 41.01 is nearly four times wine's 10.81, indicating far more severe collinearity in the directions that do exist. Eigenvalue analysis in figure 3 shows us the largest eigenvalue of 1.58 is 41 times the smallest at 0.039, meaning some encoded categorical combinations dominate the variance while many others contribute nearly nothing. Collinearity is high, both qualitatively (dummy variables summing to one) and quantitatively (many near-

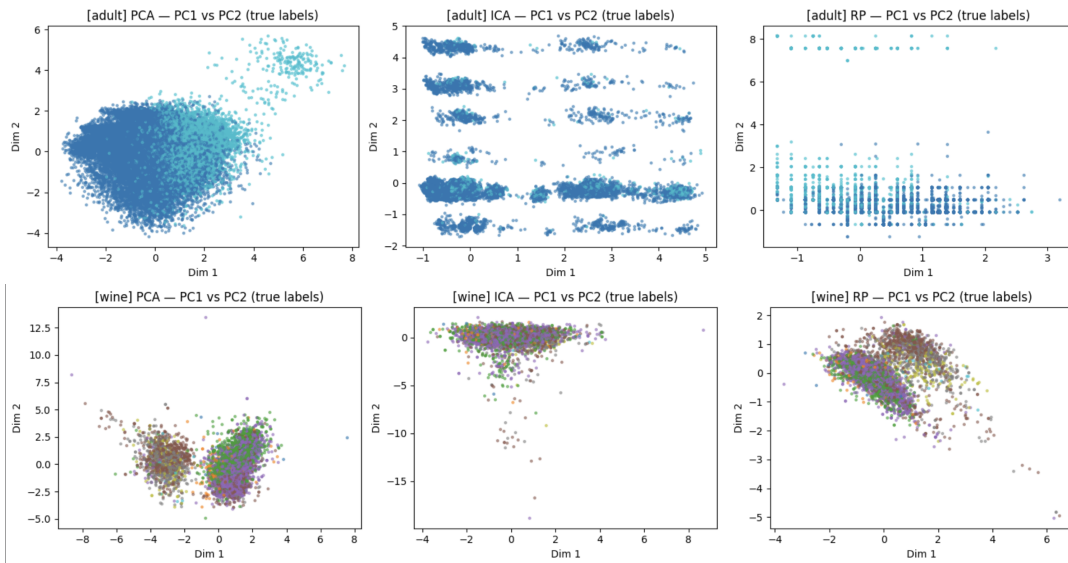


Fig. 2: Visualisation of Dimension Reduction algorithms results with true labels on Adult and Wine

zero eigenvalues). These data properties strongly influence algorithm outputs. RP and PCA mainly compress redundancy with RP reconstructing the structure effectively due to the 3:1 compression. ICA extracts categorical contrasts rather than latent sources. Noise analysis in this figure also reveals algorithm properties, at zero noise RP starts highest silhouette at 0.146, PCA at 0.124, and ICA at 0.081. ICA degrades the fastest, dropping to 0.025 at $\sigma=0.5$. This confirms the hypothesis that ICA is the most noise-sensitive method on adult. When even small amounts of noise are added the near-binary activation patterns of one-hot features become blurred. PCA degrades more gracefully, from 0.124 to 0.067, its variance-aligned components are more robust to noise. RP starts highest, eventually declining to 0.046 at $\sigma=0.5$, similar to ICA. This shows that distance preservation breaks down when noise variance approaches signal variance.

The hypothesis that PCA would perform best was confirmed on the Wine dataset. The plot shows two distinct lobes separating along Dim 1, with one dense cluster on the right and one more spread out group on the left. This is consistent with the red vs. white split identified in clustering. PCA shows a steeply decaying eigenvalue distribution with the first few principal components capturing most of the variance. PCA requires 9 out of 12 components to capture 96.7% of the variance. PCA seems to have captured the dominant axis of chemical variation in the data which corresponds to type rather than quality. Clustering will probably perform well on this space but this might not improve classification as the type rather than quality is seen in the variance.

The ICA plot shows a different structure. The bulk of the points are compressed into a dense horizontal band near $y=0$, with a downward spike of scattered points. This is a direct consequence of ICA's high kurtosis (11.89), the algorithm has found a component with a heavy-tailed sparse distribution and where most wines score near zero but a small number of outliers are in one direction. These ICA

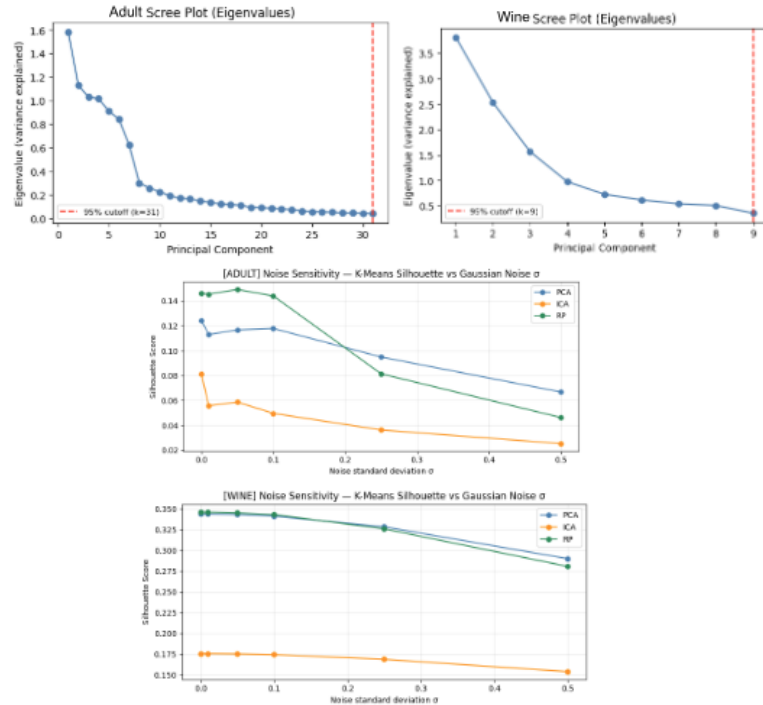


Fig. 3: Eigenvalue (PCA) and Noise Analysis (All DR) on Adult and Wine

axes seem to capture meaningful latent structure, specifically rare or extreme chemical combinations, seen in the visually elongated projections.

As predicted, Randomized Projections offered minimal benefit. There is an elongated cloud with a slight two cluster separation but with more overlap. This is expected as RP preserves pairwise distances approximately but does not align dimensions with any meaningful structure. RP also shows high instability when run across multiple seeds for K-Means

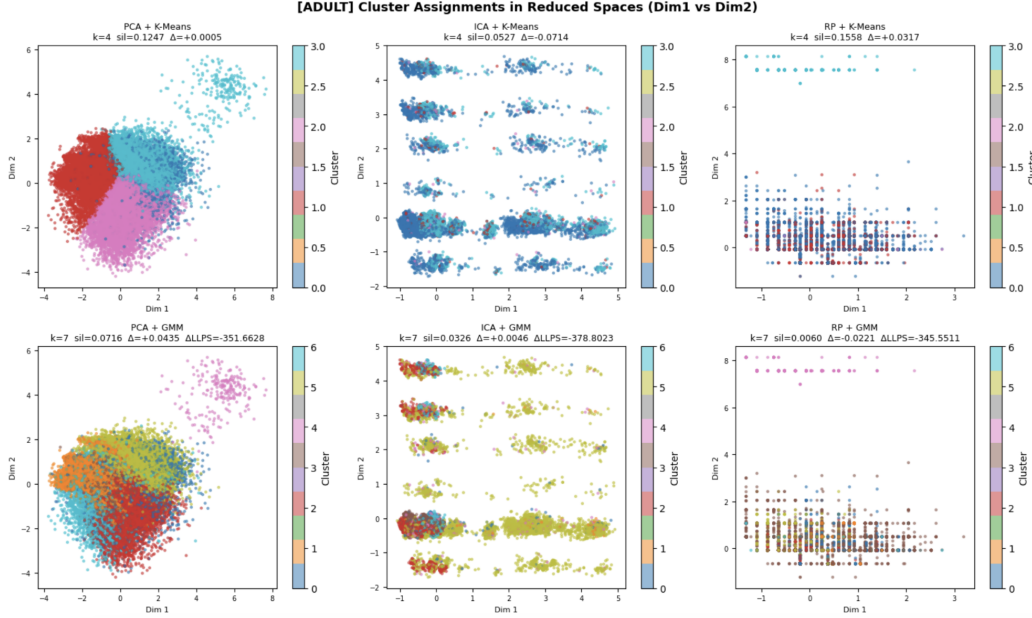


Fig. 4: Cluster Assignments on Adult for combinations of all 6 Cluster+Dimensionality-Reduction algorithms

with a standard deviation of 0.0579 (silhouette) and range of 0.1759. The scores vary largely from 0.2309 to 0.4608. 5 seeds improve K-Means baseline and 5 fall below it. With only 12 original dimensions and 9 projected dimensions, each random projection matrix has little room to average out. The Johnson-Lindenstrauss guarantee requires many more dimensions and samples before the bound becomes tight.

In the noise analysis (figure 3), all three methods degrade slowly with increase in noise on Wine and aren't too sensitive to it. PCA and RP move at nearly the same pace (from 0.344 to around 0.29 and 0.28). ICA starts and stays lower throughout the increase in noise but degrades most slowly as it's silhouette penalty comes from outlier spike rather than noise sensitivity on this dataset. Across all three methods no quality based separation is visible in first two dimensions which is consistent with the hypothesis that wine quality is too noisy and subjective to correspond to geometric cluster structure. Wine is a full-rank dataset and a rank-to-dimension ratio of 1.0, meaning no feature is an exact linear combination of others. In the eigenvalue analysis (figure 3), the top principal component alone explains 31.7% of variance and the top three together explain 65.8%. The moderate correlation drives PCA to uncover strong global structure and allow ICA to isolate hidden signals, helping both methods produce meaningful components on this dataset.

VI. CLUSTERING ON REDUCED DIMENSION SETS

On Adult, K-Means and RP is the best combination with a Silhouette of 0.1588 (+0.032 over raw) which is the largest K-Means improvement across either dataset. The scatter plot (Figure 4) shows the same grid-like pattern seen before in RP, with overlap of clusters. The four clusters show the clearest spatial separation of any adult K-Means plot with the cyan color occupying a distinct region and the remaining three

TABLE I: Step 3 clustering results in reduced spaces.

Dataset	DR	Dims	Algo	k	Sil	ΔSil	LLPS	ΔLLPS
6*Wine	PCA	9	K-Means	2	0.3447	+0.0080	—	—
	PCA	9	GMM	10	0.0554	+0.0135	-9.46	-6.13
	ICA	9	K-Means	2	0.1769	-0.1597	—	—
	ICA	9	GMM	10	0.0608	+0.0188	-9.75	-6.41
	RP	9	K-Means	2	0.2798	-0.0568	—	—
	RP	9	GMM	10	0.0096	-0.0323	-4.69	-1.35
6*Adult	PCA	31	K-Means	4	0.1247	+0.0005	—	—
	PCA	31	GMM	7	0.0716	+0.0435	19.92	-351.66
	ICA	31	K-Means	4	0.0527	-0.0714	—	—
	ICA	31	GMM	7	0.0326	+0.0046	-7.22	-378.80
	RP	31	K-Means	4	0.1558	+0.0317	—	—
	RP	31	GMM	7	0.0060	-0.0221	26.03	-345.55

colors dividing the main mass. The discrete geometry in the grid-like structure creates natural boundaries in the grid points which is why RP outperforms the continuous PCA space for K-Means. The guarantee of distance preservation (Johnson-Lindenstrauss bound) is also meaningful in this space as Euclidean distances are unreliable, compressing to 31 dimensions while preserving distance restores the geometry K-Means needs.

PCA shows almost no improvement at 0.1247 (+0.0005), the scatter plot shows four colored regions with visible boundaries and some overlap. PCA+K-Means clusters show a different arrangement than raw, the isolated population to the upper right is still present, suggesting PCA has preserved whatever feature combination made that subgroup distinct. However the remaining three clusters divide the main mass differently from raw, reflecting that PCA's rotation redistributes the variance across 31 dimensions.

ICA +K-Means is the worst result at 0.0527 (-0.071). K-Means assumes that clusters are spherical and the horizontal bands are problematic in this setting. The clusters are very different than raw since ICA has restructured the entire coor-

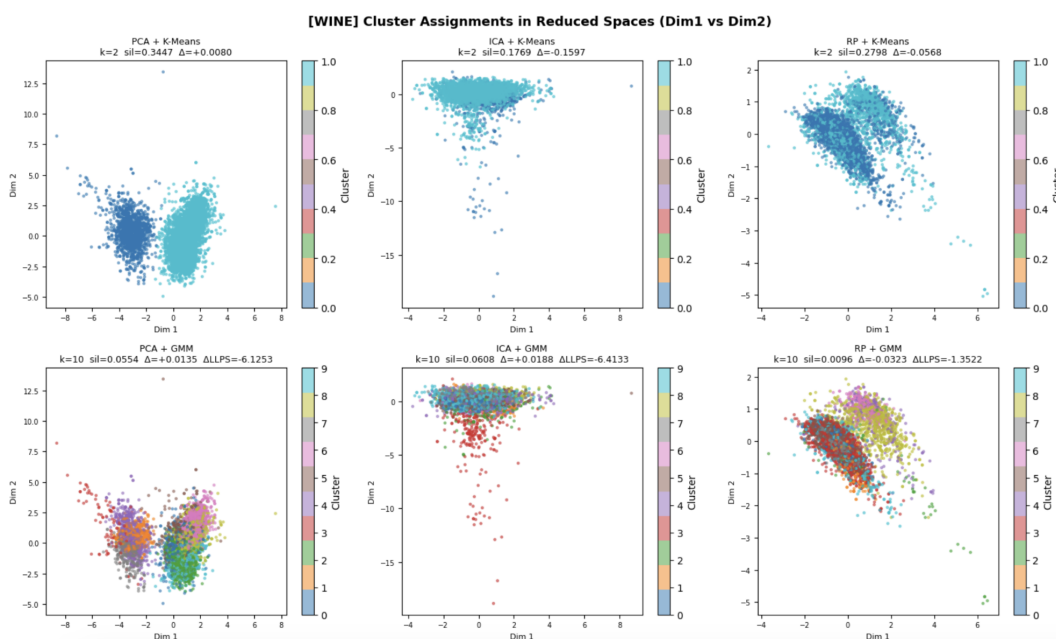


Fig. 5: Cluster Assignments on Wine for combinations of all 6 Cluster+Dimensionality-Reduction algorithms

dinate geometry. Since K-Means' spherical centroid cannot represent any individual band well, the centroid ends up floating between bands or spanning multiple bands. This results in a lower score.

The best GMM combination by silhouette is PCA + GMM at 0.0716 (+0.0435 over raw). The PCA+GMM plots show that there are fewer clean partitions than in K-Means, there is a lot of spatial overlap. The ICA+GMM (0.0326 silhouette, +0.0046 compared to raw) plot shows cluster assignments within the bands are more fragmented than K-Means. This suggests that though GMM can model elongated shapes, the non-Gaussian nature of the ICA space makes this difficult resulting in broken segments rather than a cohesive group.

RP+GMM has a silhouette of 0.0060 (-0.0221 over raw). The points in the RP+GMM plot are the most with points of different colors all over. However, this has the best LLPS (Log likelihood per sample), this means that the best model for high-dimensional sparse data is one that captures interleaved density of the population. This supports the hypothesis that the income classes don't seem to have naturally isolated clusters but overlapping distributions. The results seem to support the hypothesis that Adult dataset has a limited ability to clearly separate classes due to sparsity and imbalance. Silhouette score (Rousseeuw, 1987) [2] evaluates clustering quality by comparing intra-cluster cohesion and inter-cluster separation; the consistently low values observed (<0.16) indicate substantial overlap between clusters, supporting the conclusion that the Adult dataset lacks distinct group structure.

On Adult, the prediction that Randomized Projections (RP) would improve clustering in sparse spaces was validated. RP + K-Means achieved the highest silhouette score (0.1558), outperforming PCA. However, PCA still slightly

improves clustering on K-Means and ICA performs worst as predicted. However, although GMM models were expected to perform better due to their probabilistic nature, the silhouette scores were all lower than K-Means. GMM does have good LLPS of 26.02 for RP, this suggests that the GMM model might be capturing the complex conditional interactions mentioned in hypothesis, even if they aren't spatially distinct enough to raise the silhouette score.

On wine (Figure 5), K-Means+PCA is the best combination with a silhouette of 0.3447 (+0.008 over raw). This is the only model that improves the score over raw. The scatter plot shows two well-defined lobes cleanly divided into two colors, similar to clustering on raw. This shows that PCA's noise reduction from dropping three dimensions sharpens the chemical split without fundamentally changing it. RP + K-Means degrades to 0.2798 (-0.057) and the scatter plot reflects this, the two colored regions overlap considerably in the central mass with no clean boundary. RP produces intermediate consistency. This is consistent with RP's distance preservation guarantee: global pairwise distances are approximately maintained so the two chemical populations remain roughly separable, but the random mixing of feature directions blurs the specific boundary that PCA and raw K-Means find precisely.

ICA+K-Means is the worst result at 0.1769 (-0.160) and the scatter plot reflects this. ICA produces the most dramatically different clusters from raw on wine. Nearly the entire dataset is assigned to one cluster color, with only the small number of extreme outliers in the downward spike assigned to the second cluster. This is the strongest contradiction of the hypothesis across all twelve combinations, ICA was expected to expose useful independent chemical structure but instead completely destabilizes K-Means.

GMM once again performs worse than all K-Means results by silhouette. For GMM on wine, the best combination by silhouette is ICA+GMM at 0.0608 (+0.019 over raw). ICA + GMM achieves marginally better geometric separation by silhouette. The ICA + GMM scatter plot shows that unlike K-Means, GMM distributes its ten components more sensibly across both the main horizontal band and the outlier spike, different colors appear in the downward tail. Soft assignments allow the model to treat the spike as a mixture component without forcing a hard centroid toward it. RP+GMM has the worst silhouette score of 0.0096 (-0.032). The scatter plot showing ten components scattered randomly across the point cloud with no spatial coherence whatsoever, confirming that random projection’s mixing of the feature space is incompatible with GMM’s density estimation.

The PCA+GMM scatter plot shows ten components fragmented across the two-lobe structure with no clearly dominant separation, this is consistent with the near-zero silhouette score (0.0554, +0.0135 over raw). BIC selects $k=10$ because the data have distributional complexity, but that complexity doesn’t correspond to a good geometric clustering. Neither approaches raw GMM density fit, this confirms the hypothesis that Wine does not benefit from dimensionality reduction for GMM density modeling. Overall, the hypothesis on Wine that clustering would identify moderate structure but struggle with quality labels is supported by the binary separation in the PCA plots and moderate silhouette scores. As hypothesized, PCA was most beneficial DR method with the PCA+K-Means silhouette score being 0.3447, highest across all sets. This confirms that reducing the redundancy in features clarifies the cluster structure. It was hypothesized that ICA would be less effective due to correlated features violating independence assumptions which was confirmed on K-Means. Additionally, randomized Projections are expected to preserve distances but provide smaller benefits, this was confirmed as RP slightly degrades on both K-Means and GMM but does not have big losses either.

VII. NEURAL NETWORKS ON REDUCED DIMENSION

TABLE II: Summary Table for NN on DR methods

Rep.	Dims	Best Epoch	Best Val Loss	Train F1	Test F1
Original	12	164	1.0263	0.4954	0.3662
PCA	9	185	1.0485	0.5041	0.3615
ICA	9	157	1.0420	0.4995	0.3644
RP	9	200	1.0473	0.5009	0.3474

Overall, the results from step 4 show us that dimensionality reduction does not improve predictive performance on Wine, and the original 12-dimensional feature set produces the best F1 0.3662. All three reduced spaces underperformed the original but the margins are very small. This aligns with the hypothesis that dimensionality reduction would not meaningfully help classification on Wine since the original space is already low-dimensional and the information discarded by compression turns out to matter for prediction even if it did not matter for clustering.

Both PCA (0.3615) and ICA (0.3644) tracked the original model’s performance closely. The preservation of F1 shows that most of the discriminative information in the wine features is retained even after dimensionality reduction, particularly in PCA and ICA which aim to capture variance and statistical independence respectively. RP produced the lowest score (0.3473) which is expected as RP does not preserve structure and may distort feature relationships. In terms of generalization behavior, the reduced-dimension models show nearly identical patterns to the original. All representations show training F1-macro of around 0.50 and test F1-macro of around 0.35, this suggests similar degrees of overfitting. This implies that the neural network’s capacity dominates the generalization behavior more than differences in feature representation. Similar to F1-score none of the DR methods improve the generalization but neither did they substantially improve it. This shows that the original feature set likely contained some redundancy.

The training process differs by method. ICA shows a worse best validation loss of 1.042 compared to the original’s 1.026. This small divergence suggests that ICA’s independent components expose non-Gaussian structure that the network fits well but does not fully generalize. PCA has the worst validation loss at 1.0485 alongside the second longest training run of 185 epochs with suggests that the model is fitting the PCA space more aggressively and overfitting slightly. RP is the only representation to run the full 200-epoch. The failure to converge within the epoch limit is the most informative training behavior result in Step 4, it indicates that RP introduces noise into the feature space to prevent the network from finding a stable loss minimum. Despite RP having a higher train F1 of 0.5009 than the original, its test F1 is the lowest by a meaningful margin, revealing the worst generalization gap of any representation.

PCA and ICA offer nearly equivalent performance while reducing dimensionality by 25%, which may be beneficial for computational efficiency or model interoperability. RP showed weaker performance and offered no benefit. In summary, dimensionality reduction did not meaningfully improve predictive performance, but PCA and ICA provide compact representations that preserve accuracy, while RP appears less useful for this dataset and model.

VIII. NEURAL NETWORKS WITH CLUSTER DERIVED FEATURES

TABLE III: Summary table for NN with Cluster Features

Method	Dims	Test F1	(F1)
Original baseline	–	0.3662	–
KM one-hot only	2	0.1335	-0.2327
KM distances only	2	0.1582	-0.2080
GMM posteriors only	10	0.2393	-0.1269
Orig + KM one-hot	14	0.3695	+0.0033
Orig + KM distances	14	0.3860	+0.0198

Cluster features alone seem to be insufficient on Wine. K-Means one-hot assignments with only 2 dimensions produce a test F1 of 0.1335, and K-Means centroid distances

at 0.1582 are similarly poor, both roughly a third of the original baseline performance. GMM posteriors alone at 10 dimensions are better at 0.2393 but still are less than the baseline of 0.3662. This is expected given the clustering in step 1 seems to capture the red vs white wine split rather than quality so giving the network only two features encoding wine type provides almost no quality relevant information. When cluster features are appended to the original, the performance improves. K-Means centroid distances appended to the original is the best overall result at 0.3860, improving over baseline by 0.0198. The reason for this might be that centroid distances encode how chemically typical or atypical each wine is relative to the dominant chemical groups found in step 1. The distances appear to carry quality relevant signals that the original features don't represent. K-Means one-hot appended shows only a marginal gain of +0.0033. This makes sense as one-hot encoding tells the network only which side of the binary partition each wine falls on, while distances tell it how far resulting in smaller gain.

GMM posteriors gave a surprising result, reducing the F1-macro by -0.0133 despite adding ten soft probability features. This contradicts the hypothesis that GMM's richer probabilistic features would outperform K-Means. The likely explanation is that GMM's ten overlapping components from Step 1 has near-zero silhouette and represented continuous density variation rather than distinct chemical groups. The noisy posterior probabilities adds 10 dimensions of largely redundant information. This training F1 is 0.5502 and the highest across all representations. This shows that the model is overfitting to the GMM probability features, the validation loss is also the lowest across representations. This shows that the model clearly fails to generalize well perhaps due to the noisy GMM features. Overall, K-Means centroid distances are the most useful cluster-derived feature for wine prediction. GMM posteriors provide more information theoretically but introduce enough noise and redundancy to hurt rather than help. This is consistent that wine's main structure is a binary chemical split found in earlier steps.

IX. CONCLUSION

The experiments across all five steps largely validated the hypothesis but had a few contradictions. The main prediction on Wine was that clustering would find groups that would not be aligned with the target variable labels. This was confirmed, the dominant structure in Wine was the split between red wine and white wine rather than quality. It was hypothesized that there would not be good clustering on Adult due to high-dimensionality and sparsity. The clusters found in Adult seem to correspond to demographic subgroups rather than the true labels of income classes. Therefore, we can conclude that wine quality is subjectively rated and income arises from conditional feature interactions that cannot be captured by simple geometric partitions.

On clustering, the most clearly contradicted hypothesis on both datasets was that GMM would outperform K-Means. The reasoning was that soft probabilistic boundaries would

better handle overlapping distributions. However, this underestimated how severely high-dimensional sparse covariance degrades GMM on Adult. K-Means avoids this unstable covariance estimation, resulting in three times better silhouette. On wine, both algorithms found essentially identical binary structure when constrained to the same number of components, confirming that the apparent performance gap was an artifact of different model selection criteria rather than a genuine algorithmic difference.

Dimensionality reduction techniques behaved as predicted. On Adult, RP was most beneficial on K-Means because distance preservation directly addresses the curse of dimensionality in sparse binary spaces. PCA does not preserve the distance relationships that K-Means needs but was most beneficial to GMM. PCA eliminates the collinearity introduced by one-hot encoding, which is beneficial to GMM. ICA was the worst method for Adult on both algorithms, producing horizontal banding due to one-hot encoding that disrupts K-Means spherical assumptions.

On Wine, all three methods offered minimal benefit because the dataset is full rank, this leaves little redundancy to eliminate. PCA marginally improved K-Means by removing slight noise in the three dropped dimensions. RP was highly seed-unstable, this confirms that the Johnson-Lindenstrauss guarantee, is too loose to be reliable when compressing an already small feature space. ICA contradicted the hypothesis by revealing meaningful latent structure through high kurtosis components. This structure helped GMM rather than K-Means, as heavy-tailed sparse components are fundamentally incompatible with Euclidean distance geometry.

The neural network experiments reinforced that dimensionality reductions does not improve classification on Wine. However, K-Means centroid distances appended to the original features produced the only meaningful classification improvement across steps 4 and 5, suggesting that cluster-derived geometric signals can add predictive value when they encode real but underrepresented structure. The persistently modest performance in all steps confirms the main prediction that both datasets have fundamentally difficult learning problems that clustering or dimensionality reduction alone can't solve. Wine quality is too subjectively labeled and adult income is too conditionally structured for clustering to bridge the gap between feature space and prediction target.

AI Use Statement

ChatGPT and Google Gemini were used to generate templates for training and testing parts of the models. Additionally, after completing the report I used Google Gemini to correct spelling and grammar errors.

REFERENCES

- [1] Johnson, W. B., & Lindenstrauss, J. (1984). Extensions of Lipschitz mappings into a Hilbert space. *Contemporary Mathematics*, 26, 189–206 [Online]. Available: <https://stanford.edu/class/cs114/readings/JL-Johnson.pdf>
- [2] Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65. [Online].

Available: <https://www.sciencedirect.com/science/article/pii/S0377042787901257>